

The pricing surface: bandwidth is the unique eviction-priced dirty resource

Draft (2026-08-10). Research report for the pricing-surface track. This report restricts which local observations may affect a price; it does not change the observer transition semantics of `local-classes.md`.

Claim labels used throughout:

- **Verified** means checked against an accepted AIP, a mechanized repository fact, or the exact finite calculation in §3.
- **Desk-proved** means a complete mathematical argument is given here but is not mechanized.
- **Conjectured** means a precise plausible statement whose proof is not supplied.
- **Wall** means the stated conclusion is false under the named interpretation.

1. Verdict first

Desk-proved, headline. Let the dirty observer have

$$Q = \{C, D\}, \quad q_0 = C, \quad F = L = C, \quad W = D, \quad P = \text{id}.$$

A local price on its **positive eviction surface** is an observation-blind word price plus a nonnegative surcharge supported only on (up, D) . At every positive grade its semantic class monoid modulo word prices and exchange potentials is

$$\mathcal{P}_{\text{dirty},n}^{\text{ev}} \cong \mathbb{N}[\textit{writeback}].$$

The generator has infinite order. Equivalently, the group completion is the torsion-free rank-one group $\mathbb{Z}[\textit{writeback}]$, while the physically realizable cone selects its nonnegative ray. This holds uniformly over the whole tower, not merely at the two computed grades.

Verified, finite calculation. Restricting the table-to-edge map of `cocycle-classification.md` to the two up columns gives rank one modulo word prices at grades one and two. Its only relation is

$$[u_C] + [u_D] = [up] = 0,$$

and $[u_D]$ has infinite order. Restricting further to the oriented positive surface leaves exactly $\mathbb{N}[u_D]$. The enumerated graph sizes and cycle ranks reproduce the earlier calculation exactly.

Wall for the weaker reading. “Only up may inspect the observer” is not, by itself, enough to prove (1) as a statement about a nonnegative monoid. It also permits a surcharge on clean up. By (2),

$$[u_C] = -[u_D],$$

so the image of all nonnegative tables on the two up columns is the entire group $\mathbb{Z}[u_D]$, not one copy of $\mathbb{N}$. A pricing surface must therefore be oriented: it specifies nonnegative **surcharges above an observation-blind baseline**, not merely a set of columns allowed to vary. The cache

model supplies the orientation—clean eviction is the free baseline and dirty eviction carries the liability—so this is a physical restriction, not a choice made to repair the algebra after the fact.

Desk-proved, scope. Under eviction-only pricing, warmth disappears but visit counters survive. For the warmth observer every state-sensitive up column is a word price plus a potential, at every grade. For the modulo- r visit observer, charging one chosen post-fill state on up represents the same exact order- r class as charging the corresponding visit on down. Thus “eviction-visible” is narrower than “nontrivial local resource,” but it does not mean “physically a write-back”: finite-state phase can also be read at a boundary.

2. Pricing surfaces

2.1 Positive surcharge surfaces

Fix a finite observer $A = (Q, q_0, F, W, L, P)$ and the generator alphabet Σ of `local-classes.md`. A local table is a function

$$T : \Sigma \times Q \rightarrow \mathbb{N}.$$

Definition. A *pricing surface* is a subset $S \subseteq \Sigma \times Q$. Its cone of admissible tables is

$$\begin{aligned} Tab_S = \{T \mid \text{there are } \lambda \in \mathbb{N}^\Sigma \text{ and } s \in \mathbb{N}^{\Sigma \times Q} \text{ such that} \\ T(a, q) = \lambda_a + s(a, q), \quad s(a, q) = 0 \text{ for } (a, q) \notin S\}. \end{aligned}$$

The vector λ is the observation-blind baseline and s is the surface surcharge. In particular, every entry off S equals the baseline for its generator and is observation-blind. The positivity in (3) is load-bearing: a surface says which observed conditions may add a charge, not which arbitrary signed deviations are allowed.

Let ρ_n be the reachable table-to-edge map and let q_n quotient edge cochains by word prices and exchange potentials, as in `cocycle-classification.md` §2. Define the fixed-grade surface image

$$\mathcal{P}_{A,n}^S = q_n \rho_n (Tab_S).$$

Unreachable surface entries map to zero. Definition (4) therefore retains the earlier reachable-image discipline: choosing a syntactic pair does not manufacture a semantic class.

There are two related objects which must not be conflated:

1. The formal-difference surface group uses $\mathbb{Z}^S$ and records all signed combinations of surface columns.
2. The physical surface monoid is the image of the cone (3). It may be a proper cone in that group, or it may contain units when distinct allowed surcharges become opposite classes.

The weak “allowed columns” reading forgets item 2 and is exactly what causes the counterexample in §1.

2.2 The eviction surface

For an observer with baseline mark q_0 , define the maximal oriented eviction surface

$$S_{\text{ev}}(A) = \{(up, q) : q \neq q_0\}.$$

Thus non-baseline observer state may add a charge only when the focused subtree is left. The distinguished baseline entry $T(up, q_0) = \lambda_{up}$ pins the orientation. A model may choose a smaller liability set $E \subseteq Q \setminus \{q_0\}$ and use $\{up\} \times E$.

For dirty, $q_0 = C$ and (5) is the singleton

$$S_{\text{ev}}(\text{dirty}) = \{(up, D)\}.$$

Consequently every admissible dirty table has the unique pointwise normal form

$$T(a, q) = \lambda_a + k \mathbf{1}_{(a,q)=(up,D)}, \quad \lambda \in \mathbb{N}^\Sigma, \quad k \in \mathbb{N}.$$

This is already the “word prices plus write-back” existence statement. The content of the uniqueness theorem is that semantic quotienting introduces no identification between different values of k .

3. Exact grades one and two

3.1 Calculation

The finite check repeats `cocycle-classification.md` §2.4 with only the surface columns retained.

1. Enumerate the reachable augmented graph from every base memory and legal initial head position, with all observer marks initially q_0 .
2. Keep parallel addressed-generator edges and choose an undirected spanning forest.
3. Form fundamental-cycle residual columns for the six word prices and the allowed table entries.
4. Compute the exact integer relations among those columns. Rational rank detects the free surface directions; the integral relations and the closed-walk witness detect torsion and the positive image.

Verified, finite computation. For dirty the results are:

grade	V	E	b_1	rank of word columns	rank after u_C, u_D	surface group
1	20	72	53	5	6	$\mathbb{Z}$
2	272	992	721	5	6	$\mathbb{Z}$

In the column order

$$(down0, down1, up, read, write0, write1, u_C, u_D),$$

the integral kernel is generated at both grades by

$$(-1, -1, 1, 0, 0, 0, 0, 0) \quad \text{and} \quad (-1, -1, 0, 0, 0, 0, 1, 1).$$

The first vector is the depth coboundary $down0 + down1 - up = \delta(depth)$. Subtracting it from the second gives (2). There is no further relation, so $[u_D]$ is a primitive infinite-order generator and the restricted group is torsion-free. Intersecting with the cone (6) permits only $k[u_D]$ for $k \geq 0$, proving (1) at the two finite grades. Allowing both up states instead gives $a[u_C] + b[u_D] = (b - a)[u_D]$, whose positive image is all of $\mathbb{Z}$.

3.2 Direct fixed-grade proof

The computation has a short conceptual certificate.

Theorem, desk-proved (dirty uniqueness at a fixed grade). For every $n \geq 1$, the map

$$\mathbb{N} \rightarrow \mathcal{P}_{\text{dirty}, n}^{\text{ev}}, \quad k \mapsto k[u_D]$$

is an isomorphism of commutative monoids.

Proof. Surjectivity is the normal form (7), because the quotient removes λ . For injectivity use the words `shortA n` and `shortB n` from `Adic.Dirty`. They start at the same configuration, end at the same configuration, and have identical counts of every generator. Their clean instruction costs are therefore equal. The first has no dirty up; the second has exactly n . If $k[u_D] = k'[u_D]$, exchange on these two paths cancels both the potential and all word prices and gives

$$kn = k'n.$$

Since $n \geq 1$, $k = k'$. $\square$

The same witness shows that no positive multiple of $[u_D]$ vanishes, so the group completion of (9) is torsion-free. Grade zero has no runnable up and is intentionally excluded: its surface image is zero.

4. The tower

Let $D = \Sigma \times Q$ be the grade-independent table domain and let $\mathcal{R}_n \subseteq \mathbb{Z}^D \oplus \mathbb{Z}^\Sigma$ be the relation lattice of `cocycle-classification.md` equation (20). For a surface S , let i_S include its formal surcharge columns and the baseline word-price columns into that fixed presentation space. Then

$$\mathcal{R}_n^S = i_S^{-1}(\mathcal{R}_n).$$

Desk-proved (surface restriction preserves descent). Whenever the strong-connectivity theorem gives $\mathcal{R}_{n+1} \subseteq \mathcal{R}_n$, taking preimages in (10) gives

$$\mathcal{R}_{n+1}^S \subseteq \mathcal{R}_n^S.$$

Thus imposing a pricing surface only shrinks the relation lattice. The rational surface spaces still form a descending chain in one fixed finite-dimensional space and eventually stabilize. No new saturation claim is needed or made.

Desk-proved, dirty tower theorem. Dirty is strongly connected at every grade by `cocycle-classification.md` §5.5, so (11) applies. More strongly, the normal form (7) and the witness proof of (9) are uniform in n . Hence the entire positive-grade tower has the same class monoid

$\mathbb{N}[\textit{writeback}]$, with the same coefficient k in every grade. This direct proof does not depend on an effective stabilization bound and does not touch the report's integral saturation wall.

5. Which resources are eviction-visible?

5.1 Warmth vanishes

For warmth, $Q = \{C, H\}$, $q_0 = C$, $F = H$, $P = C$, and $W = L = \text{id}$. The maximal oriented surface (5) permits only the hot-up surcharge u_H .

Desk-proved. Let $A_C(x)$ be the number of cold nodes on the active head path. A cold active node can only come from the arbitrary initial head position: entering a node applies F and makes it hot, while `ParentFill` resets only inactive children. Therefore a cold up consumes exactly one unit of A_C , and no other step changes A_C . After complementing this bounded finite-grade potential, u_C is an exchange coboundary. Pointwise,

$$u_C + u_H = u_P,$$

so u_H is a word price plus a potential as well. Every warmth eviction surface table is trivial modulo word prices at every grade.

Verified, finite computation. At grades one and two, adding both u_C and u_H raises the word-column rank by zero. The exact relations contain $u_C = 0$ and $u_H = \textit{down0} + \textit{down1}$ modulo potentials. The graph triples $(|V|, |E|, b_1)$ are respectively $(40, 128, 89)$ and $(3248, 10080, 6833)$, reproducing the prior warmth enumeration.

Warmth's original nontrivial witness is necessarily down-priced: it detects whether a fill finds a cold or hot child. Once only the leaving boundary may inspect state, that information has been overwritten by $F = H$.

5.2 The modulo- r counter survives

For the visit observer, $Q = \mathbb{Z}/r\mathbb{Z}$, $q_0 = 0$, $F(q) = q + 1$, and $W = L = P = \text{id}$. Let u_j charge an up whose source mark is j .

Desk-proved. The down into a node observes $j - 1$, changes it to j , and the matching up observes j ; the mark cannot change while that node remains on the active path. The discrepancy on open paths is the boundary of the number of active nodes in state j . Hence u_j is exchange-equivalent to charging the corresponding phase transition on down. Phase-edge charges on the cyclic state graph differ by vertex potentials, so every $[u_j]$ is the same generator.

The closed word

$$(\textit{down0up})^r$$

visits every phase once and has u_j -cost one. Its word price is a multiple of r , so $k[u_j] \neq 0$ for $1 \leq k < r$. Conversely, the node-counter potential from `local-classes.md` §4.5 shows that $r[u_j]$ is one static down price. Thus $[u_j]$ has exact order r at every positive grade. Since the oriented eviction surface contains u_j for every $j \neq 0$, its positive image is the cyclic group $\mathbb{Z}/r$ as soon as $r \geq 2$.

Verified, finite computation for $r = 2$. At grades one and two, the surface columns add no rational rank, but the primitive integral relation is $2u_j = \text{down}0 + \text{down}1$ modulo potentials. The graph triples are (48, 160, 113) and (7168, 24576, 17409), matching the prior parity calculation. The two-visit loop proves that the order is two rather than one.

5.3 Summary

observer	original priced event	eviction-surface image	tower verdict	reading
dirty	dirty up	$\mathbb{N}[\text{writeback}]$	uniform for $n \geq 1$	unique positive bandwidth resource
warmth	cold down	0	uniform	invalidation is fill-visible, not eviction-visible
visit mod r	wrapping down	$\mathbb{Z}/r$	uniform for $n \geq 1$	finite phase is boundary-visible

The table separates two questions. A class may be visible at eviction without being bandwidth, as the counter shows. The uniqueness theorem says that for the **dirty observer** and its physical positive surface, bandwidth is the only non-word-price direction.

6. Physical reading

The surface is the algebraic form of an existing machine decision. AIP-2 identifies down with acquisition and clean up with free discard; cache-v0.md §4b's discharged call 2 identifies up with eviction and makes every dirty eviction pay its eager write-back. AIP-5 fixes the same dirty observer and explains flush-at-boundary as a gauge change rather than a second operational semantics. Equation (6) merely says which event is allowed to inspect the liability bit under those decisions: potential- consuming clean departure remains at the blind baseline, while a dirty departure externalizes pending data and adds a nonnegative charge.

7. Lean statement ladder

The present deliverable is a report, not a mechanization. The following statements fit the existing Adic.Dyadic.Dirty vocabulary and reuse Config, step, run, Prices, letterCost, Exchange, shortA, and shortB.

First separate the named surcharge from the already mechanized total dirty cost:

```
def writebackCharge (operation : LocalOp) (config : Config n) : Nat :=
  match operation, config.marks with
  | .up, true :: _ => 1
  | _, _ => 0

def writebackCost : Word → Config n → Nat
```

```

| [], _ => 0
| operation :: word, config =>
  match step operation config with
  | none => 0
  | some next =>
    writebackCharge operation config + writebackCost word next

def evictionSurfaceCost (prices : Prices) (coefficient : Nat) :
  Word → Config n → Nat := fun word config =>
  letterCost prices word config + coefficient * writebackCost word config

```

The structural and compatibility statements should then be:

```

theorem writebackCost_isCocycle :
  IsCocycle (writebackCost (n := n))

theorem dirtyCost_eq_clean_add_writeback
  (word : Word) (config final : Config n)
  (hrun : run word config = some final) :
  dirtyCost word config = cleanCost word + writebackCost word config

theorem writebackCost_shortA (n : Nat) :
  writebackCost (shortA n) (root n) = 0

theorem writebackCost_shortB (n : Nat) :
  writebackCost (shortB n) (root n) = n

```

The headline uniqueness statement can avoid defining an abstract quotient:

```

theorem evictionCoefficient_unique
  (n : Nat) (hn : 1 ≤ n)
  (leftPrices rightPrices : Prices) (left right : Nat)
  (potential : Config n → Nat)
  (h : Exchange
    (evictionSurfaceCost (n := n) leftPrices left)
    (evictionSurfaceCost (n := n) rightPrices right)
    potential) :
  left = right

```

Its proof is the existing `dirty_not_cohomologous` proof with `shortA` and `shortB`, retaining the coefficient instead of specializing it to one. The surface normal form is most naturally encoded as data rather than as a predicate on arbitrary tables:

```

structure DirtyEvictionTable where
  prices : Prices
  writeback : Nat

def DirtyEvictionTable.cost (table : DirtyEvictionTable) :
  Word → Config n → Nat :=
  evictionSurfaceCost table.prices table.writeback

```

```

theorem dirtyEvictionTable_classification
  (n : Nat) (hn : 1 ≤ n) :
  Function.Injective (fun k : Nat =>
    -- semantic exchange class of coefficient k at grade n
    dirtyEvictionClass n k)

```

The last theorem needs the repository’s eventual concrete class-quotient type; until then `evictionCoefficient_unique` is the subtraction-free, fully meaningful theorem. It states exactly that the semantic surface monoid is a free commutative monoid on write-back.

8. Boundaries and friction

- The result is relative to the dirty observer and the oriented positive surface. It does not classify arbitrary finite observers.
- The unrestricted up-observable cone is not an $\mathbb{N}$ -ray; clean-up surcharge supplies the negative generator. Any future use of “eviction surface” must retain the baseline/orientation in definition (3).
- The modulo- r result is a mathematical boundary-visible class, not a claim that a visit counter is physical bandwidth.
- Surface restriction inherits the rational descending-chain theorem but does not solve or use the general irreversible-observer saturation wall.
- The exact calculation was straightforward once the earlier table-to-edge map was reconstructed. The non-obvious point was that positivity lives in the presentation cone, not in the SNF group completion; omitting that distinction reverses the headline monoid.

References inside the project

- AIP-2, `aips/accepted/0002-dyadic-machine-v0.md`, especially the free clean up amendment.
- AIP-5, `aips/accepted/0005-dirty-cocycle.md`, especially §§1, 2, and 5.
- `aips/draft/cache-v0.md` §4b, discharged dirty-up pricing call.
- `aips/draft/local-classes.md`, observer semantics and fixed-grade exchange criterion.
- `aips/draft/cocycle-classification.md` §§2, 3.2, and 5, table-to-edge SNF, failed unrestricted theorem, and local-to-global descent.
- `lean/Adic/Dirty.lean`, mechanized dirty configuration, costs, witnesses, and `dirty_not_cohomologous`.